

Python Fundamentals

Lab 3

Description: This lab covers functions and while loops in Python.

Deadline: 17 December 2025

Submission Instructions : Rename your file with your name for example (MohamedHamed.pdf, MohamedHamed.py or MohamedHamed.ipynb) and submit it via this form: [Python Task3](#)

Create a simple ATM / Bank application in Python with the following requirements:

✧ Functions to implement:

- deposit() – Allows the user to deposit money into their account
- withdraw() – Allows the user to withdraw money from their account
- show_balance() – Displays the current account balance

✧ Main function: Include a while loop that repeatedly shows the user the available options:

- 1) Deposit
- 2) Withdraw
- 3) Show Balance
- 4) Exit

Prompt the user to enter their choice and stop the loop if the user chooses Exit

✧ Additional Requirements:

- Validate user inputs (e.g., cannot withdraw more than the available balance)
- Maintain an account balance variable that is updated with deposits and withdrawals
- Use clear messages for successful transactions and errors

✧ Example Flow:

```
Welcome to ATM
Choose an option:
1. Deposit
2. Withdraw
3. Show Balance
4. Exit
Enter choice: 1
Enter amount to deposit: 500
Deposit successful! Your new balance is 500.

Choose an option:
...
```